Informatica® Cloud Data Integration

Microsoft Fabric Data Warehouse Connector

Informatica Cloud Data Integration Microsoft Fabric Data Warehouse Connector
October 2026

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Publication Date: 2026-10-15

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Preface

Use Microsoft Fabric Data Warehouse Connector to learn how to read or write to Microsoft Fabric Data Warehouse by using Data Integration. Learn to create a Microsoft Fabric Data Warehouse connection, develop, and run mappings and mappings in advanced mode, and tasks in Data Integration.

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CHAPTER 1

Introduction to Microsoft Fabric Data Warehouse Connector

You can use Microsoft Fabric Data Warehouse Connector to securely read data from and write data to Microsoft Fabric Data Warehouse from Data Integration.

Create a Microsoft Fabric Data Warehouse connection and use the connection as a source, target, or lookup in mappings and mapping tasks.

You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality. A mapping in advanced mode can run on an advanced cluster hosted on Microsoft Azure or a self-service cluster.

You can also create a mapping in SQL ELT mode to perform the data transformation entirely within your cloud ecosystem.

Microsoft Fabric Data Warehouse Connector assets

Create assets in Data Integration to integrate data using Microsoft Fabric Data Warehouse Connector.

When you use Microsoft Fabric Data Warehouse Connector, you can include the following Data Integration assets:

- Mapping
- Mapping - SQL ELT. For more information on mappings in SQL ELT mode, see Chapter 4, “Mappings in SQL ELT mode for Microsoft Fabric Data Warehouse” on page 26.
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for Microsoft Fabric Data Warehouse

Create a Microsoft Fabric Data Warehouse connection to securely read data from or write data to Microsoft Fabric Data Warehouse.

Before you create a connection, enable the connector for the runtime environment. For more information about enabling connectors, see *Service and connector assignment* in the Administrator help.

You can use a Microsoft Fabric Data Warehouse connection to specify sources and targets in mappings and mapping tasks.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Power BI prerequisites:
   a. Create a workspace, if it does not already exist.
   b. Create a lakehouse in the workspace. Make sure you have the required admin privileges to create a lakehouse.

   For more information about creating a workspace and lakehouse, see Creating a workspace and lakehouse.
2. Microsoft Azure portal prerequisites:
   a. Register your application in Microsoft Entra ID and generate client ID, tenant ID, and client secret. Microsoft Fabric Data Warehouse Connector does not require any API permissions assigned to your application. Additionally, you can remove any API permissions that are assigned to the application.
   b. Create a security group with assigned owners or members in Microsoft Entra ID.
   For more information about creating a security group, see Creating a security group.
3. To allow the Microsoft Entra app to use the Power BI APIs, enable the Power BI service admin settings, and then add the Microsoft Entra app to the workspace in Power BI.

   Note: The Manage access pane in the Microsoft Entra app allows you to add roles based on the required capabilities. The pane displays admin, member, contributor, and viewer roles. The maximum number of capabilities are associated with the admin role. However, to access Microsoft Fabric OneLake and perform read and write operations, you need at least the Contributor role.

   For more information about enabling access in the Power BI admin portal and adding the Microsoft Entra app to the workspace, see Enabling access in the Power BI portal.

Connect to Microsoft Fabric Data Warehouse

Let's configure the Microsoft Fabric Data Warehouse connection properties to connect to Microsoft Fabric Data Warehouse.

Before you begin

Before you get started, you'll need to keep the client ID, client secret, workspace name, database name, and tenant ID details handy.

Check out "Prerequisites" on page 8 to make sure that the setup is complete and all the prerequisites are in place before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, or serverless runtime environment. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
SQL Connection String	The SQL connection string to connect to Microsoft Fabric Data Warehouse. Specify the connection string in the following format: `<Server>.datawarehouse.pbidedicated.windows.net`
Client ID	The application ID or client ID of your application registered in Azure Active Directory for service principal authentication.
Client Secret	The client secret for your application registered in Azure Active Directory.
Tenant ID	The tenant ID of your application registered in Azure Active Directory.

Property	Description
Workspace	The name of the workspace in Microsoft Fabric Data Warehouse that you want to connect.
Database	The name of the database in Microsoft Fabric Data Warehouse that you want to connect.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Schema Name	The name of the schema in Microsoft Fabric Data Warehouse where the tables are stored.

Additional Properties	Additional connection properties that you configure in key-value pairs.
	You can configure the following properties:
	- connectRetryCount. The maximum number of attempts to establish or re-establish a connection.
	You can configure an integer between 0 and 255. A value of 0 indicates that a retry won't be attempted.
	Default is 3.
	- connectRetryInterval. The time, in seconds, between connection retry attempts.
	You can configure an integer between 1 and 60. This property is ignored if connectRetryCount is set to 0.
	Default is 10 seconds.
	- loginTimeout. The number of seconds to wait for a connection to be established.
	Connection attempts continue up to the limit defined by connectRetryCount. However, if the total retry time exceeds the loginTimeout, any attempt to establish the connection stops when the loginTimeout duration is reached.
	Maximum is 65535 seconds. Default is 30 seconds.
	- WorkspaceId. The ID of the workspace that you want to connect to in Microsoft Fabric Data Warehouse.
	- LakehouseId. The ID of the lakehouse used to stage data.
	Configure this property when you enable the Copy command to stage data as Parquet files and load staging data to Microsoft Fabric Warehouse.
	Enter the property in the following format:
	`LakehouseId=<ID>`
	For example, in the following URL, the lakehouse ID is the value that follows `/lakehouse/`:
	`https://app.fabric.microsoft.com/workloads/lakehouse/<lakehouse-ID>`
	Note: Ensure that the lakehouse ID that you specify in the connection is valid.

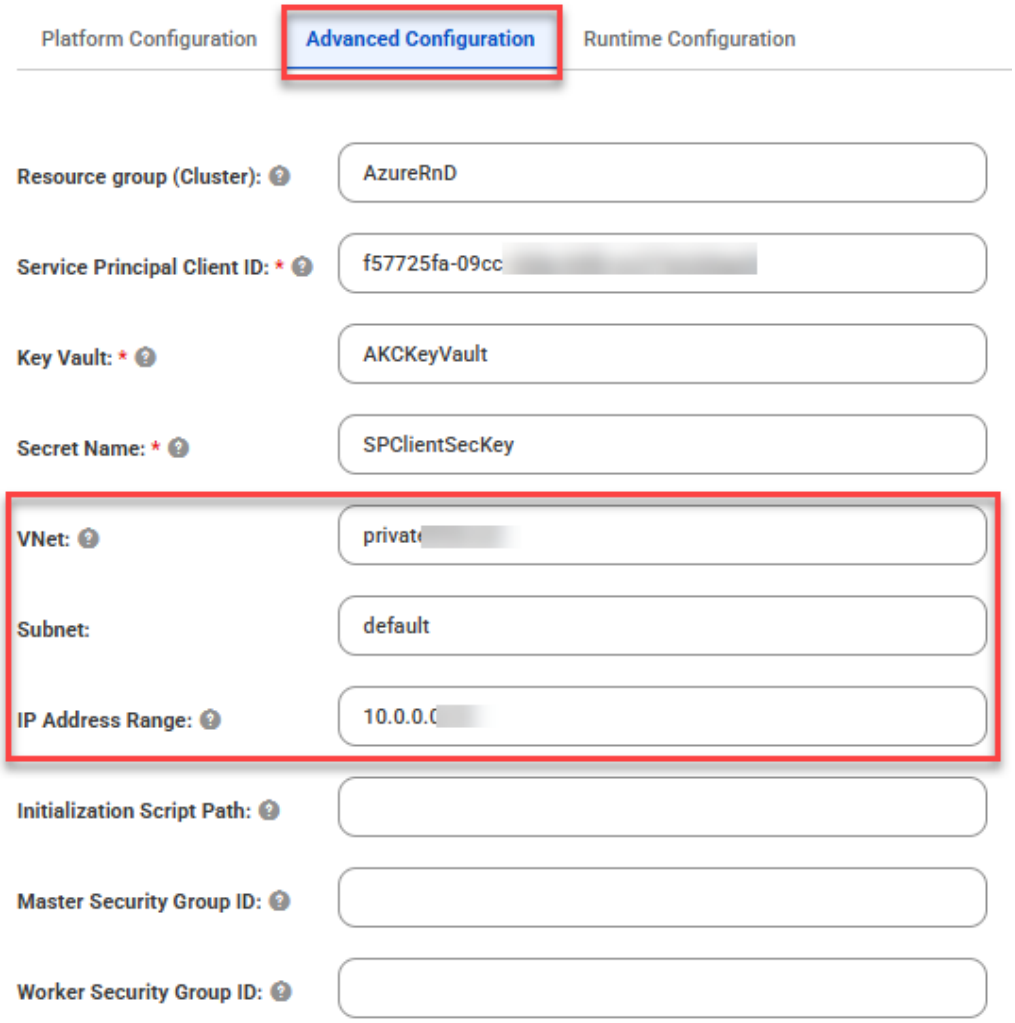
Private links to access Microsoft Fabric Data Warehouse

You can access Microsoft Fabric Data Warehouse using Azure Private Link endpoints.

To connect to the Microsoft Fabric Data Warehouse account over the private Azure network, see Secure connectivity to Microsoft Fabric.

To run mappings in advanced mode over the private Azure network, additionally you must perform the following steps after you complete the required prerequisites:

1. Log in to Intelligent Data Management Cloud (IDMC), and then click **Administrator**.
2. In **Administrator**, navigate to **Advanced Clusters** to create or modify an Advanced Configuration.
3. In the **Advanced Configuration** tab, add the Vnet, Subnet, and IP Address Range properties that is linked with the private DNS zone. This helps to spawn the cluster resources under this subnet and provide access to the private DNS zone.



CHAPTER 3

Mappings and mapping tasks with Microsoft Fabric Data Warehouse Connector

You can configure a mapping to read from and write to Microsoft Fabric Data Warehouse tables.

You can configure a Source, Target, or Lookup transformation to represent a Microsoft Fabric Data Warehouse object. Use the Mapping Designer in Data Integration to add the Source, Target, or Lookup transformations in the mapping canvas and configure the source, target, and lookup properties. To create a mapping, select **Mapping** as the mapping type in the Mapping Designer.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality. When you run a mapping in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

You can also create a mapping in SQL ELT mode when your source and target are in the same cloud ecosystem and you want to perform the data transformation entirely within the cloud ecosystem. For more information about mappings in SQL ELT mode, see Chapter 4, “Mappings in SQL ELT mode for Microsoft Fabric Data Warehouse” on page 26.

Microsoft Fabric Data Warehouse sources in mappings

To read data from Microsoft Fabric Data Warehouse, configure a Microsoft Fabric Data Warehouse object as a source in a mapping or mapping in advanced mode.

Specify the name and description of the Microsoft Fabric Data Warehouse source. Configure the source and advanced properties for the source object.

The following table describes the source properties that you can configure for a Microsoft Fabric Data Warehouse source:

Property	Description
Connection	Name of the source connection. Select an existing connection, create a new connection, or define parameter values for the source connection property. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format. Specify the parameter file directory and name in the advanced session properties. When you switch between a non-parameterized and a parameterized Microsoft Fabric Data Warehouse connection, the advanced property values are retained.
Source Type	Type of the source object. Select Single Object, Query, or Parameter. Default is Single Object. **Note:** You cannot read multiple Microsoft Fabric Data Warehouse objects, even though Multiple Objects appear in the list of source objects.
Parameter	A parameter file where you define values that you want to update without having to edit the task. The **Parameter** property appears only if you select the source type as **Parameter**. Select an existing parameter for the source object or click **New Parameter** to define a new parameter for the source object. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format.
Object	Name of the Microsoft Fabric Data Warehouse source object. Select the source object for a single source. You can also select an existing object from the list. Ensure that the table headers of the source object do not contain special characters.

The following table describes the query options that you can configure for a Microsoft Fabric Data Warehouse source:

Property	Description
Filter	Filters record based on the filter condition in a read operation. Click **Configure** to add conditions to filter records from the data that the Secure Agent reads from the source. You can specify the following filter conditions: - Not parameterized. Uses a basic filter to specify the object, field, operator, and value to select specific records. - Completely parameterized. Uses a parameter to represent the field mapping. - Advanced. Uses an advanced filter to define a more complex filter condition. Default is Not parameterized.
Sort	Sorts records based on the conditions you specify. You can specify the following sort conditions: - Not parameterized. Allows you to select the fields and type of sorting to use. - Parameterized. Uses a parameter to specify the sort condition. - Sort Order. Determines whether to sort data in an ascending or descending order.

The following table describes the advanced source properties that you can configure for a Microsoft Fabric Data Warehouse source:

Property	Description
Schema Name Override	Overrides the schema name configured in the source object and uses the specified schema name to read data at runtime.
Table Name Override	Overrides the table name configured in the source object and uses the specified table name to read data at runtime.
SQL Override	The SQL statement to override the default query that is used to read data from the Microsoft Fabric Data Warehouse source.
Pre SQL	The pre-SQL command to run on the Microsoft Fabric Data Warehouse source table before the Secure Agent reads the data. For example, if you want to update records in the database before you read the records from the table, specify a pre-SQL statement.
Post SQL	The post-SQL command to run on the Microsoft Fabric Data Warehouse table after the Secure Agent completes the read operation. For example, if you want to delete some records after the latest records are loaded, specify a post-SQL statement.
On Pre-Post SQL Error	Determines the behavior when a task that includes pre-SQL or post-SQL commands encounters errors. Select one of the following options: - **Continue**. The task continues regardless of errors. - **Stop**. The task stops when errors occur while running pre-SQL or post-SQL commands.
Tracing Level	Sets the amount of detail that appears in the log file. Select one of the following options from the list: - Terse - Normal - Verbose Initialization - Verbose Data Default is Normal.

Key range partitioning

You can configure key range partitioning when you use a mapping task to read data from Microsoft Fabric Data Warehouse sources.

When you run a mapping configured with key range partitioning, the Secure Agent distributes rows of source data based on the fields that you define as partition keys. The Secure Agent compares the field value to the range values for each partition and sends rows to the appropriate partitions.

Use key range partitioning for columns that have an even distribution of data values. Otherwise, the partitions might have unequal size.

For example, a column might have 10 rows between key values 1 and 1000 and the column might have 999 rows between key values 1001 and 2000. If the mapping includes multiple sources, use the same number of key ranges for each source.

When you define key range partitioning for a column, the Secure Agent reads the rows that are within the specified partition range.

For example, if you configure two partitions for a column with the ranges as 10 through 20 and 30 through 40, the Secure Agent does not read the rows 20 through 30 because these rows are not within the specified partition range.

You can configure a partition key for fields of any numeric data type. However, you cannot use decimals in key range values.

Key range partitions are not available in the following scenarios:

- When you create a mapping in advanced mode.
- When the mapping includes any of the following transformations:
  - Web Services
  - XML to Relational
- When you use custom queries and SQL overrides in the mapping.

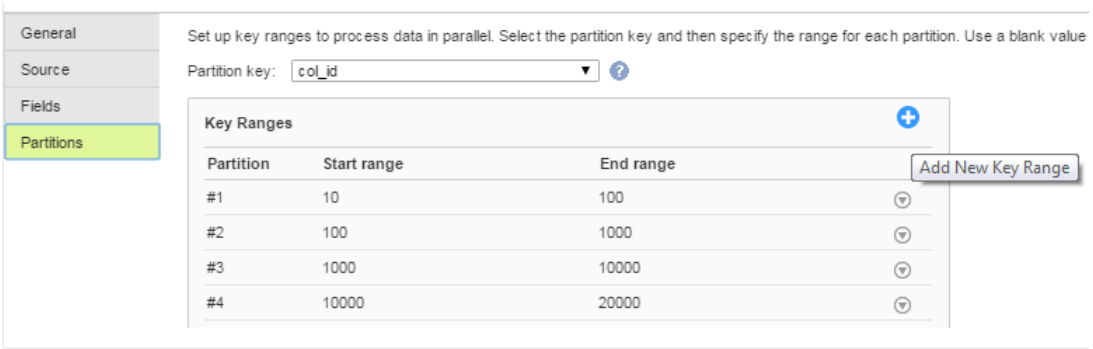
Configuring key range partitioning

Perform the following steps to configure key range partitioning for Microsoft Fabric Data Warehouse sources:

1. In the Source properties, click the **Partitions** tab.
2. Select the required partition key from the list.
3. Click **Add New Key Range** to define the number of partitions and the key ranges based on which the Secure Agent must partition data.

   Use a blank value for the start range to indicate the minimum value. Use a blank value for the end range to indicate the maximum value.

   The following image displays the **Partitions** tab:



IDENTITY column

You can read data from Microsoft Fabric Data Warehouse tables that contain an IDENTITY column. The IDENTITY column appears as a metadata column on the Fields tab of the object properties.

Consider the following guidelines when you use the IDENTITY column:

- IDENTITY columns apply to read operations only. You can't write to an IDENTITY column.
- You can use IDENTITY columns in key range partitioning in the Source transformation.
- An IDENTITY column must be of the Bigint data type.
- You can't define a seed or increment value for an IDENTITY column, so you can't reseed the column.
- IDENTITY column values are positive and unique, but they aren't sequential.

- You can't enable dynamic schema change handling for the IDENTITY column.

Microsoft Fabric Data Warehouse targets in mappings

To write data to a Microsoft Fabric Data Warehouse, configure a Microsoft Fabric Data Warehouse object as a target in a mapping or mapping in advanced mode.

Specify the name and description of the Microsoft Fabric Data Warehouse target. Configure the target and advanced properties for the target object.

The following table describes the target properties that you can configure for a Microsoft Fabric Data Warehouse target:

Property	Description
Connection	Name of the target connection. Select an existing connection, create a new connection, or define parameter values for the target connection property. If you want to overwrite the target connection properties at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format. Specify the parameter file directory and name in the advanced session properties. When you switch between a non-parameterized and a parameterized Microsoft Fabric Data Warehouse connection, the advanced property values are retained.
Target Type	Type of the Microsoft Fabric Data Warehouse target object. Select one of the following source types from the list: - Single Object. Select to specify a single Microsoft Fabric Data Warehouse object. - Parameter. Select to specify a parameter name. You can configure the target object in a mapping task associated with a mapping or mapping in advanced mode that uses this Target transformation. Default is Single Object.
Parameter	A parameter file where you define values that you want to update without having to edit the task. The **Parameter** property appears only if you select the target type as **Parameter**. Select an existing parameter for the target object or click **New Parameter** to define a new parameter for the target object. If you want to overwrite the target connection properties at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. Ensure that the parameter file is in the correct format.
Object	Name of the Microsoft Fabric Data Warehouse target object. Specify the target object that you want to create. You can also select an existing object from the list.

Property	Description
Operation	Type of target operation. Select Insert, Update, Delete, or Data Driven. Default is Insert. You cannot perform data driven target operation in a mapping. However, you can perform data driven target operation in a mapping in advanced mode.
Data Driven Condition	The expression to flag rows for an insert, update, or delete operation when you select data driven as the operation.
Update Column	The temporary key column that identifies rows in the Microsoft Fabric Data Warehouse target table to update or delete data. **Note:** This property appears only if you select delete, update, or data driven operation in the **Operation** property.

The following table describes the advanced target properties that you can configure for a Microsoft Fabric Data Warehouse target:

Property	Description
Update Mode	Loads data to the target based on the mode you select. This property is applicable when you select the **Update** target operation. Select one of the following modes: - Update As Update. Updates all rows flagged for update if the entries exist. - Update Else Insert. The agent first updates all rows flagged for update if the entries exist in the target. If the entries do not exist, the agent inserts the entries.
Copy Command	Uses the Copy command to load data from the staging location to Microsoft Fabric Data Warehouse. The Copy command performs a bulk load. It stages all rows into a single Parquet file and loads them in one operation, rather than row by row. Before you use the Copy command, you must enter the lakehouse ID in the additional properties of the connection or the Target transformation. **Note:** Ensure that the lakehouse ID that you specify in the additional properties of the connection or the Target transformation is valid. If you enter the lakehouse ID in both the connection and the Target transformation, the value in the Target transformation takes precedence. When you use the Copy command, ensure that the data type, precision, and scale of the target columns are compatible with the incoming data. **Note:** You can't use the Copy command in mappings in advanced mode.
Schema Name Override	Overrides the schema name configured in the target object and uses the specified schema name to write data at runtime.
Table Name Override	Overrides the table name configured in the target object and uses the specified table name to write data at runtime.
Staging Lakehouse Name	The name of the lakehouse where the data is staged in a temporary table before the data is written to the target.
Truncate Table	Truncates the target before inserting data to the target.

Property	Description
Pre SQL	The pre-SQL command to run on the Microsoft Fabric Data Warehouse target table before the Secure Agent writes the data.
Post SQL	The post-SQL command to run on the Microsoft Fabric Data Warehouse target table after the Secure Agent writes the data.
On Pre-Post SQL Error	Determines the behavior when a task that includes pre-SQL or post-SQL commands encounters errors. Select one of the following options: - Continue. The task continues regardless of errors. - Stop. The task stops when errors occur while running pre-SQL or post-SQL commands.
Update Override	Overrides the default update query that the agent generates for the update operation. This property applies when you select the target operation as update or when you use the data driven operation for rows flagged for an update.

Additional Properties	Additional runtime properties that you want to configure when you write to Microsoft Fabric Data Warehouse.
	You can configure the following properties:
	- MaxNumberOfRetries. The maximum number of retries for metadata refresh and database operations.
	Default is 6.
	- RetryDelayInMillis. The delay in milliseconds between the retries in case of failure.
	Default is 10000 milliseconds.
	- FabricSQLEndpointId. The ID of the SQL endpoint in the lakehouse where data is stored temporarily before the data is written to the target.
	- LakehouseId. The ID of the lakehouse used to stage data.
	Configure this property when you enable the Copy command to stage data as Parquet files and load staging data to Microsoft Fabric Warehouse.
	Enter the property in the following format:
	`LakehouseId=<ID>`
	For example, in the following URL, the lakehouse ID is the value that follows `/lakehouse/`:
	`https://app.fabric.microsoft.com/workloads/lakehouse/<lakehouse-ID>`
	To add multiple properties, enter each property on a new line.

Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the mapping task forwards rejected rows to the next transformation.

Microsoft Fabric Data Warehouse lookups in mappings

Use a Microsoft Fabric Data Warehouse Lookup transformation to look up data in a Microsoft Fabric Data Warehouse object and retrieve data based on a specified lookup condition.

You can use the Microsoft Fabric Data Warehouse Lookup transformation to look up the customer name in another Microsoft Fabric Data Warehouse object. For example, the source table includes the customer code, but you want to include the customer name in the target table to make summary data easy to read.

You can create the following lookups when you configure field mappings in a mapping task:

- Connected with cached or uncached
- Unconnected with cached

You cannot configure an uncached lookup for mappings in advanced mode.

The following table describes the Microsoft Fabric Data Warehouse lookup object properties that you can configure in a Lookup transformation:

Property	Description
Connection	Name of the lookup connection. You can select an existing connection, create a new connection, or define parameter values for the lookup connection property. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. You can switch between a non-parameterized and a parameterized Microsoft Fabric Data Warehouse connection. When you switch between the connections, the advanced property values are retained.
Source Type	Type of the source object. Select Single Object, Query, or Parameter.
Lookup Object	Name of the lookup object for the mapping.
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the lookup object or click **New Parameter** to define a new parameter for the lookup object. The **Parameter** property appears only if you select parameter as the source type. If you want to overwrite the parameter at runtime, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties.
Define Query	The query to look up data when the selected source type is a query. If the source type is a query, click **Edit Custom Query**, enter a valid custom query, and then click **OK**.

Property	Description
Multiple Matches	Behavior when the lookup condition returns multiple matches. You can select from the following options in the lookup object properties to determine the behavior: - Return first row[1] - Return last row[1] - Return any row - Return all rows - Report error
[1]Doesn't apply to mappings in advanced mode.	

The following table describes the Microsoft Fabric Data Warehouse lookup advanced properties that you can configure in a Lookup transformation:

Property	Description
Schema Name Override	Overrides the schema specified in the connection.
Table Name Override	Overrides the table name of the imported Microsoft Fabric Data Warehouse source table.
SQL Override	Overrides the default SQL query used to read data from the Microsoft Fabric Data Warehouse source.
Pre-SQL	Pre-SQL command to run before reading data from the source.
Post-SQL	Post-SQL command to run after reading data from the source.
On Pre-Post SQL Error	Determines the behavior when a task that includes pre-SQL or post-SQL commands encounters errors. You can select any of the following options: - Continue. The task continues regardless of errors. - Stop. The task stops when errors occur while running pre-SQL or post-SQL commands.
Tracing Level[1]	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.
Lookup Data Filter[1]	Limits the number of lookups that the mapping performs on the cache of the lookup source table based on the value you specify in the filter condition. Maximum length of the filter condition is 32768 characters. This property is applicable when you select object as the source type and enable lookup cache on the **Advanced** tab for the Lookup transformation. For more information about this property, see *Transformations* in the Data Integration documentation.
[1]Doesn't apply to mappings in advanced mode.	

Process stored procedures using an SQL transformation

You can configure an SQL transformation to process stored procedures in a Microsoft Fabric Data Warehouse mapping. When you add an SQL transformation, you define the database connection and the type of SQL that the transformation processes.

You cannot configure an SQL transformation for mappings in SQL ELT mode and mappings in advanced mode.

The stored procedure must exist in the Microsoft Fabric Data Warehouse before you create the SQL transformation. When the SQL transformation processes a stored procedure, it passes input parameters to the stored procedure. The stored procedure returns the values to the output fields of the transformation.

For more information about stored procedures, see *Transformations* in the Data Integration documentation.

Rules and guidelines for SQL transformations

Consider the following rules and guidelines when you configure an SQL transformation in mappings to process a stored procedure:

- When you use a parameterized connection in an SQL transformation, you can process only a stored procedure in a Microsoft Fabric Data Warehouse mapping. The other options SQL query and stored function are not applicable.
- The decimal and numeric data types are rounded off to the nearest higher integer. For example, the decimal boundary value 9999999999.99999 is rounded to 10000000000.0000.
- The data exceeding 4000 characters in the Char, Varbinary, and Varchar data types is truncated even though Data Integration can handle up to 8000 characters.
- When you map an incoming field to an output field in a stored procedure, and the stored procedure doesn't return a value for an output field, the mapping returns the value of the incoming field instead of null. This is a third-party limitation.

Dynamic schema handling

When you add a mapping to a mapping task, you can choose how Data Integration handles changes in the data object schemas. To refresh the schema every time the task runs, you can enable dynamic schema handling in the task.

A schema change includes one or more of the following changes to the data object:

- Fields added, deleted, or renamed.
- Fields updated for data type, precision, or scale.

Configure schema change handling in the **Advanced Options** section on the **Runtime Options** tab when you configure the task.

The following table describes the schema change handling options:

Option	Description
Asynchronous	Default. Data Integration refreshes the schema when you edit the mapping or mapping task, and when Intelligent Data Management Cloud (IDMC) is upgraded.
Dynamic	Data Integration refreshes the schema every time the task runs. You can choose from the following options to refresh the schema: - **Alter and apply changes.** Data Integration alters the target schema and adds the new fields from the source. - **Don't apply DDL changes.** Data Integration does not apply the schema changes to the target. - **Drop current and recreate.** Data Integration drops the existing target table and then recreates the target table at runtime using all the incoming metadata fields from the source.

For more information about dynamic schema handling, see *Tasks* in the Data Integration documentation.

Rules and guidelines for dynamic schema handling

Consider the following rules and guidelines when you enable dynamic schema change handling in a mapping:

- You cannot make changes to the columns of the Uniqueidentifier data type.
- When you read data from Microsoft Fabric Data Warehouse, any table or column constraints in the source table are not retained in the target.
- When you add a new column to the source data, Data Integration determines the most suitable data type for the target based on the data type of the new source column.
  For example, the Uniqueidentifier data type in the source data is mapped to Varchar data type in the target or the Date data type in the source data is mapped to the Datetime2 data type in the target.
- Special characters in the column name are written as underscore in the target.
- When you select the **Alter and apply changes** option, you cannot refresh schema from a stored procedure. Additionally, Data Integration only adds the new fields from the source and does not rename fields, delete fields, or apply precision and scale updates.
- When you create a new target at runtime and select the **Don't apply DDL changes** option, schema changes must not be applied. However, Data Integration applies the schema changes to the new target. When you create a new target subsequently and select the **Don't apply DDL changes** option, the schema changes are not applied to the new target.
- You cannot enable dynamic schema handling to refresh the schema when you parameterize a connection or an object.

Rules and guidelines for mappings

Consider the following rules and guidelines for Microsoft Fabric Data Warehouse mappings:

General guidelines

- When you configure a filter in a mapping that reads data, consider the following guidelines:
  - If the table or schema name contains Unicode characters, the mapping fails. To resolve this issue, enclose the table name or schema name with Unicode characters in square brackets within the filter condition.

- If the column name starts with a number, the mapping fails.
- If you prefix the column name with an underscore in the filter condition, the mapping fails. You need to enter the actual column name in the filter condition.
- If the mapping contains an advanced filter for the object, for example, <Table2>, in the query options section and an override to the table name in the advanced properties, use the table name that you specify as an override in the advanced properties directly in the advanced filter condition. For example, <Table2>.id >= 1

- If the column or schema name in a custom query, an SQL override, or Update override field contains Unicode characters, enclose the column name or schema name with Unicode characters in square brackets.
- Ensure that the query you specify for the pre and post SQL statements are valid. Otherwise, the mapping fails.
- When you read from Microsoft Azure Synapse SQL and write to Microsoft Fabric Data Warehouse, the mapping fails.
- When you specify a custom query for a source object, ensure that the query does not contain a semicolon at the end of the SQL statement. Otherwise, the mapping fails.
- You cannot use input parameters in custom queries that read data from Microsoft Fabric Data Warehouse source objects.
- Mapping fails with an Invalid Temp_table_Object error when the staging Lakehouse is schema enabled.
- When you stop a job or mapping from the My Jobs page, temporary tables in the staging Lakehouse are not automatically deleted.
- When you specify a single constant with the IIF condition in the data driven condition such as IIF(col_bigint>747654,DD_INSERT), the Secure Agent inserts the data into the target even for those rows that do not satisfy the condition.
- When you specify the DD_REJECT constant in the data driven condition, the Secure Agent does not log the rejected rows in the error file or the session log.
- When you configure a parameterized mapping with the **Create New at Runtime** option, enter the schema name/table name in the **Object** field in the Target transformation. The mapping validates the entry and throws a validation error if a table with the same name exists. However, if you enter only the table name and a table with the same name exists, the mapping does not validate the entry and inserts data to the existing table.
- A mapping that uses a MINUS operator in a custom query to read data from Microsoft Fabric Data Warehouse source objects fails at runtime. Use an EXCEPT operator instead of a MINUS operator.
- When you use a custom query in the Source transformation to call a stored procedure, the mapping fails at runtime.
- When you configure key range partitioning in the Source transformation in a mapping that reads data which contains null values, the mapping ignores the null values and does not write them to the target.
- When you sort records in the Source transformation, you cannot use key range partitioning. Even if you configure key range partitioning when you sort records, it is not considered.
- To avoid a Java heap size error when you write to Microsoft Fabric Data Warehouse, you need to allocate more memory to the DTM in the Secure Agent properties based on the amount of data that you want to write. Increase the -Xms and -Xmx values for the DTM in the JVM options in the system configuration details of the Secure Agent. The recommended -Xms value is 512 MB, while the recommended -Xmx value is 1024 MB.

- When you write to Microsoft Fabric Data Warehouse, the mapping fails if the table name or schema name contains unsupported special characters. However, the table name or schema name can contain the following special characters:

  - !,@,#,$,%,^,&,_,|,:,",?,>,<
- When you filter records in a read operation, you can use only the following filter operators:

  - Equals
  - Greater
  - Greater_or_equals
  - Less_or_equals
  - Less
  - Not_Equals
- Even though you cannot read from multiple Microsoft Fabric Data Warehouse objects, the Multiple Objects option appears in the Source Type property. When you configure a Source transformation to read from multiple Microsoft Fabric Data Warehouse objects, the mapping fails with the following error:

  ```
  [ERROR] Failed to run the Microsoft Fabric Data Warehouse mapping because of the following error: [Invalid column name 'c_smallint'.]
  ```

Lookup Transformation

- You cannot set the default column value for the return field in a cached Lookup transformation in a Microsoft Fabric Data Warehouse mapping in advanced mode.
- You cannot use Binary and Varbinary datatype columns as lookup conditions.
- When you configure a lookup data filter in a cached lookup, you cannot use a custom query as a lookup object or configure an SQL override. However, the table name override is considered when you configure a lookup data filter.
- You cannot use a custom query as a lookup object or configure an SQL override in an uncached lookup.
- If you parameterize the lookup condition, you can only use the AND operator between multiple conditions or expressions.

SQL override

- When you specify an SQL override statement to override the default SQL query that reads data from a Microsoft Fabric Data Warehouse source, ensure that the SQL query does not contain a semicolon at the end of the SQL statement. Otherwise, the mapping in advanced mode fails.
- When you use an ORDER BY clause along with Join or Union clauses in a custom query or in an SQL override statement to read data from Microsoft Fabric Data Warehouse source objects, ensure that you add "offset 0 rows" at the end of the query. Otherwise, the mapping fails.
- If the mapping contains both a filter for the source object and an SQL override statement in the advanced source properties, the query in the SQL override statement takes precedence over the filter condition.
- When you specify an SQL override statement in the Source transformation to call a stored procedure in Microsoft Fabric Data Warehouse, the mapping in advanced mode fails at runtime.
- The count and order of columns in an SQL override must match the count and order of columns in the connected fields.

Update override

- Specify the update query in the following format:

```
UPDATE <Target table name> SET <Target table name>.<Column1> = :TU.<Column1>,
<Target table name>.<Column2> = :TU.<Column2>, … <Target table name>.<ColumnN>
= :TU.<ColumnN> FROM :TU WHERE <Target table name>.<Update Column1> = :TU.<Update
Column1> AND <Target table name>.<Update Column2> = :TU.<Update Column2> AND …
<Target table name>.<Update ColumnN> = :TU.<Update ColumnN>
```

- Column names for :TU must match the target table column names.
- You must specify the update query with a valid SQL syntax because Microsoft Fabric Data Warehouse Connector replaces :TU with a temporary table name and does not validate the update query.
- You cannot use unconnected fields to the target in the update query.
- When you select Update Override in the Microsoft Fabric Data Warehouse target properties to override the default update SQL statement that the Secure Agent generates and also enable alter and apply changes to refresh the schema in the task, the task does not alter the target schema or add new fields from the source.

Data types

- When you write data to a Microsoft Fabric Data Warehouse target created at runtime, Float or Real data types are written as Double data type in the target. To write Float or Real data types, use the **Edit Metadata** option to edit the data type in the Target transformation.
- When you read data of the Float or Real data type from a source and write to a target, the mapping might not accurately write the Float or Real data type to the target.
  For more information about writing the Float or Real data type to Microsoft Fabric Data Warehouse, see Float and Real (Transact-SQL) in the Microsoft documentation.

Parameterization

- When you use input parameters, specify the parameter name in the following format in a mapping task: $name$.
- You can't parameterize a target created at runtime using the parameter file.
- You can't use input parameters in a parameter file to parameterize the source filter and advanced properties at runtime.
- Ensure that you create the parameter before you override it using a parameter file.
- You can't parameterize only the connection. You can either parameterize both the connection and the object or only the object.
- When you parameterize the object, ensure that the metadata of the imported object matches the metadata fetched in the override.
- When you parameterize the connection or object, ensure that the connection or object that you want to override exists in Microsoft Fabric Data Warehouse.
- When you parameterize the connection and objects using the parameter file, ensure that the object name in the parameter file does not contain the schema name. If the connection in the parameter file doesn't contain the schema name, the mapping considers the schema name configured in the mapping task. To override the schema name, use the schema override property in the advanced properties.

CHAPTER 4

Mappings in SQL ELT mode for Microsoft Fabric Data Warehouse

Create a mapping in SQL ELT mode when you want all the mapping logic to be processed by your cloud ecosystem.

You can run a mapping in SQL ELT mode to load data from the following data sources to Microsoft Fabric Data Warehouse and perform all of the data transformation within Microsoft Fabric Data Warehouse:

- Microsoft Azure Data Lake Storage Gen2
- Microsoft Fabric Data Warehouse
- Microsoft Fabric Lakehouse

When you run a mapping in SQL ELT mode, the transformation logic is translated into ecosystem-specific commands and SQL statements that run in the underlying cloud infrastructure. Because data isn't moved out of the cloud infrastructure to transform it, the overall processing speed increases.

You can also preview data for individual transformations to validate the mapping logic.

For more information, see Mappings in SQL ELT mode in *Mappings* in the Data Integration documentation.

Example

You work in the sales and inventory department of a retail organization that operates multiple stores and an online platform. The management wants to optimize its inventory management, understand customer behavior, and improve sales strategies by analyzing data from various sources.

The organization uses Microsoft Azure Data Lake Storage Gen2 to store the data from multiple sources. To analyze the data, the organization plans to leverage the warehouse infrastructure of Microsoft Fabric Data Warehouse and load all its data to Microsoft Fabric Data Warehouse.

You can use SQL ELT to quickly load data from Microsoft Azure Data Lake Storage Gen2 to Microsoft Fabric Data Warehouse. SQL ELT is faster than the ETL method and eliminates any data egress costs as the data remains within your cloud infrastructure.

Mapping configuration in SQL ELT mode

You can use the Mapping - SQL ELT mapping type to directly create a mapping in SQL ELT mode.

You can run a mapping in SQL ELT mode to load data from the following data sources to Microsoft Fabric Data Warehouse:

- Microsoft Fabric Lakehouse
- Microsoft Fabric Data Warehouse
- Microsoft Azure Data Lake Storage Gen2

To create a mapping in SQL ELT mode, you create a mapping, and then select **Mapping - SQL ELT** as the mapping type. You're then prompted to choose a Microsoft Fabric Data Warehouse target connection. If your organization doesn't have any Microsoft Fabric Data Warehouse connections, you're prompted to create one.

After you choose the target connection, the Mapping Designer opens in SQL ELT mode and you can continue to configure the required transformations.

Before you run a mapping that is configured for SQL ELT mode, you can also preview SQL queries in the SQL ELT Query panel in the Mapping Designer. After you configure a mapping and run the preview, Data Integration displays the SQL queries to be executed and any warnings in the **SQL ELT Query** panel.

For information about running an SQL ELT query preview, see Running an SQL ELT query preview for mappings in SQL ELT mode in *Mappings* in the Data Integration documentation.

For more information about configuring a mapping in SQL ELT mode, see Creating a mapping in SQL ELT mode in *Mappings* in the Data Integration documentation.

Transformations with Microsoft Fabric Data Warehouse

The Source, Target, and other transformations that you configure in mappings in SQL ELT mode can behave differently than in regular mappings.

A mapping in SQL ELT mode includes only transformations that the cloud ecosystem can process.

Some properties in mappings might not apply for mappings in SQL ELT mode. This topic lists the applicable transformations, properties in source and target transformations, functions, and operators that you can use in mappings in SQL ELT mode. It also provides rules and guidelines for specific configurations and points out where they differ from mappings when applicable.

Sources in mappings in SQL ELT mode

When you configure a source connection in a Source transformation, you can choose only a Microsoft Fabric Lakehouse or a Microsoft Fabric Data Warehouse connection.

Microsoft Fabric Data Warehouse source properties

You can configure the following properties for a Microsoft Fabric Data Warehouse source:

- Source type - Single object, query, or parameter.
- Query options - Filter. You can use completely parameterized and advanced filter conditions.

- Schema Name Override
- Table Name Override
- SQL Override

For more information about the source properties, see "Microsoft Fabric Data Warehouse sources in mappings" on page 12.

Microsoft Fabric Lakehouse source properties

You can configure the following properties for a Microsoft Fabric Lakehouse source:

- Source Type - Single object, query, or parameter.
- Query options - Filter. You can use completely parameterized and advanced filter conditions.
- Table Name Override
- SQL Override

For information about how to configure these properties, see the Microsoft Fabric Lakehouse Connector documentation.

Microsoft Azure Data Lake Storage Gen2 source properties

You can configure the following properties for a Microsoft Azure Data Lake Storage Gen2 source:

- Source Type - Single object or parameter.
- Source Type - File and Directory
- Filesystem Name Override
- Allow Wildcard Characters
- Directory Override
- File Name Override
- Recursive Directory Read
- Compression Format

For information about how to configure these properties, see the Microsoft Azure Data Lake Storage Gen2 Connector documentation.

Targets in mappings in SQL ELT mode

When you configure a Target transformation in a mapping in SQL ELT mode, you need to use only a Microsoft Fabric Data Warehouse connection.

You can configure the following properties in a Microsoft Fabric Data Warehouse target transformation in a mapping in SQL ELT mode:

- Target Type - Single Object, Parameter
- Operation - Insert, Update, Delete
- Update Mode - Update as Update, Update else Insert
- Schema Name Override
- Table Name Override
- Truncate Table
- Pre SQL
- Post SQL

- On Pre-Post SQL Error
- Update Override

For more information about the target properties, see "Microsoft Fabric Data Warehouse targets in mappings" on page 16.

Transformations in mappings in SQL ELT mode

A mapping in SQL ELT mode includes transformations that Microsoft Fabric Data Warehouse can process.

You can use the following transformations in a mapping in SQL ELT mode:

- Aggregator
- Expression
- Filter
- Joiner
- Lookup
- Rank
- Router
- Source
- Target
- Union

Functions in mappings in SQL ELT mode

When you create expressions in a mapping in SQL ELT mode, you must use the native functions and expression syntax of Microsoft Fabric Data Warehouse and not Informatica functions and expression syntax.

You can use the following native functions in a mapping in SQL ELT mode:

Aggregate functions

APPROX_COUNT()	MIN()
AVG()	STDEV()
CHECKSUM_AGG()	STDEVP()
COUNT()	SUM()
COUNT_BIG()	VAR()
GROUPING()	VARP()
MAX()	

Bit manipulation functions

- BIT_COUNT()
- GET_BIT()
- LEFT_SHIFT()
- RIGHT_SHIFT()
- SET_BIT()

Cryptographic functions

- HASHBYTES()

Date and Time functions

- GETUTCDATE()

Expression functions

- NULLIF()

JSON functions

- ISJSON()

Logical functions

- IIF()

Mathematical functions

ABS()
ACOS()
ASIN()
ATAN()
ATN2()
CEILING()
COS()
COT()
DEGREES()
EXP()
FLOOR()
LOG()
LOG10()
PI()
POWER()
RADIANS()
ROUND()
SIGN()
SIN()
SQRT()
SQUARE()
TAN()

Metadata functions

OBJECTPROPERTY()
OBJECTPROPERTYEX()
SCHEMA_ID()
SCHEMA_NAME()
STATS_DATE()

Security functions

SUSER_NAME()
SUSER_SNAME()
USER_NAME()

String functions

ASCII()
BASE64_DECODE()
BASE64_ENCODE()
CHAR()
CHARINDEX()
CONCAT()
CONCAT_WS()
DIFFERENCE()
FORMAT()
LEFT()
LEN()
LOWER()
LTRIM()
PATINDEX()
QUOTENAME()
REPLACE()
REPLICATE()
REVERSE()
RIGHT()
RTRIM()
SOUNDEX()
SPACE()
STR()
STUFF()
SUBSTRING()
TRIM()
UNICODE()
UPPER()

System functions

COMPRESS()
DECOMPRESS()
ISNULL()
ISNUMERIC()

Window functions

AVG()
COUNT()
COUNT_BIG()
CUME_DIST()
DENSE_RANK()
FIRST_VALUE()
LAG()
LAST_VALUE()
LEAD()
MAX()
MIN()
RANK()
ROW_NUMBER()
STDEV()
STDEVP()
SUM()
VAR()
VARP()

For more information on functions and guidelines, see SQL database functions in the Microsoft documentation.

Operators in mappings in SQL ELT mode

When you use mappings in SQL ELT mode, Data Integration converts the expression in the transformation by determining equivalent operators in the database. If there is no equivalent operator, Data Integration processes the transformation logic.

The table lists the operators that you can push to Microsoft Fabric Data Warehouse:

+	%	<>	AND
-	>	>=	OR
*	<	<=	NOT
/	=	!=	-

Edit field metadata in mappings in SQL ELT mode

You can edit the precision and scale of a field in the source or lookup object in a mapping that reads from Microsoft Azure Data Lake Storage Gen2.

To edit the metadata, use the **Edit Metadata** option for the source and lookup fields on the **Fields** tab of the **Properties** panel.

You can edit the following metadata based on the data type of the field:

- Decimal fields: Precision and scale
- String and binary fields: Precision

Rules and guidelines in mappings in SQL ELT mode

Certain rules and guidelines apply to mappings in SQL ELT mode.

General rules and guidelines

- When you run a mapping in SQL ELT mode that uses a custom query in the Source or Target transformation along with a table name or schema name override, the table or schema specified in the custom query takes precedence.
- When you use a Joiner transformation to join two tables with the same name and a schema name override is configured for both the sources, the mapping fails to honor the schema override.
- When you specify columns of bit or binary data types in an Aggregator transformation, make sure to add these columns to the group by field. Otherwise, the mapping fails with a runtime error. This is because the Aggregator transformation adds the MAX() function for the columns that are not added to the aggregate function or group by field.
- When you configure a Lookup transformation to look up data of the Bit data type, the mapping fails with the following error:
  `[ERROR] ELT failed due to the following error\: [Operand data type bit is invalid for max operator.]`

  Ensure that the lookup object does not contain the Bit data type.
- You cannot specify columns of Binary data type in a lookup condition.
- You cannot use Expression macros in transformations.
- You cannot use a Filter transformation to filter data with the date or datetime2 data types.
- When you use an ISNUMERIC() function, you cannot process date and datetime2 data types.
- When you use the ABS() function to process a BIGINT value of -9,223,372,036,854,775,808, the mapping fails with an arithmetic overflow error. Ensure that the lower boundary value for BIGINT is -9,223,372,036,854,775,807.
- When you use a Window function, ensure that the start offset is either 0 or a negative value and end offset is either 0 or a positive value.
- You cannot parameterize a Window function in a mapping task.
- In String functions that accept varbinary arguments, do not pass constant values of the varbinary data type. Instead, pass the varbinary column to the String function. For example, instead of `SUBSTRING(0x496E666F726D61746963612049E646961, 1, 4)` or `SUBSTRING('0x496E666F726D61746963612049E646961', 1, 4)`, you need to pass the varbinary data type in the format `SUBSTRING(col_varbin, 1, 4)` where `col_varbin` is a column of the varbinary data type.
- When you use constant values in functions in an Expression transformation, and those expressions are implicitly converted to the Bigint data type, the mapping fails with the following error:
  `[ERROR] ELT failed due to the following error\: [Invalid length parameter passed to the substring function.]`

  For more information, see Constant folding and expression evaluation in Microsoft documentation.
- When you preview data or run a mapping in SQL ELT mode to write float or real data types, the source data and target data don't match in some cases. The values written to the target are approximate and not

the same as in the source. To write the exact values, map real and float data types with numeric or decimal data types.

- When you configure input parameters while you parameterize a mapping in SQL ELT mode, do not select the **Allow parameter to be overridden at run time** option.
- When you read data from Microsoft Fabric Lakehouse or Microsoft Fabric Data Warehouse, ensure that the source, target, and lookup belong to the same workspace.
- You can only use SELECT statements in custom queries.
- When you use a custom query in the Source or Target transformation along with a SQL override, use the following format:
  - <Database name>.<schema>.<tablename> in Microsoft Fabric Data Warehouse
  - <Database-name>.dbo.<tablename> in Microsoft Fabric Lakehouse

Microsoft Azure Data Lake Storage Gen2 source guidelines

Certain rules and guidelines apply to mappings in SQL ELT mode when you load data from Microsoft Azure Data Lake Storage Gen2 source to Microsoft Fabric Data Warehouse.

- You can only configure the Shared Key authentication type to access Microsoft Azure Data Lake Storage Gen2.
- You can use only Delimiter, Qualifier, and First Data Row formatting options to read flat files.
- When you select the source type as **File**, consider the following guidelines:
  - Ensure that the **File Name Override** and **Directory Override** properties do not contain wildcard characters.
  - Do not select **Allow Wildcard Characters** and **Recursive Directory Read** options.
- When you select the source type as **Directory**, consider the following guidelines:
  - All the source files in the directory must contain the same metadata.
  - All the files must have the same file format. For example, flat file or Parquet.
  - All the files must have the same compression format to read data from the source. For example, Gzip or None.
  - Flat files in the directory must have the same formatting options such as, Delimiter, Qualifier, and First Data Row.
  - Consider the following guidelines when you specify a file in the **File Name Override** property:
    - The file specified at runtime overrides the file specified in the object.
    - The recursive directory read option is not honored.
  - You can only use the asterisk mark (*) as a wildcard character in file name and directory override.
- If files or directories contain the asterisk mark (*), the Secure Agent treats * as a wildcard character.
- Do not specify a root directory (/) when you override the directory.
- Dates before 1582-10-15 00:00:00.000000 with datetime2 datatype stored in Parquet files as int96 might result in incorrect data due to historical calendar differences.
- Select **Gzip** compression format to read Gzip compressed flat files or Parquet files.
- Select **None** compression format to read uncompressed flat files and uncompressed or Snappy compressed Parquet files.
- Mapping fails if the filename, directory name, or subdirectory name contains Unicode characters or special characters other than an underscore (_) character.
- When you edit the metadata for the decimal data type, the decimal value is rounded off to the nearest integer if the digit immediately following it is 5 or greater. For example, 3.25 is rounded off to 3.3.

CHAPTER 5

Migrating a mapping

You can configure a connection and mapping in one environment and then migrate and run the mapping in another environment.

After the migration, you can change the connection properties from the Administrator service, but you do not need to modify the mapping. Data Integration uses the configured runtime attributes from the earlier environment to run the mapping successfully in the new environment.

Migrate a mapping within the same path

If you want the migrated mapping to use the same object path as in the earlier environment, you must maintain the same schema and table in the Microsoft Fabric Data Warehouse account for both the earlier and the new environment.

For example, if you have two different accounts, Account1 used for Organization 1 and Account2 used for Organization 2, the object path for the schema and table must be the same in both the accounts:

Account1: DB1/SCHEMA1/TABLE1

Account2: DB1/SCHEMA1/TABLE1

You do not need to override the schema and table in the advanced properties.

Migrate a mapping to a different path

You can use a different object path to run the mapping from the new environment.

Before you migrate the mapping, you can change the object metadata, runtime attributes, or the connection attributes to reflect the object path in the migrated environment. You do not have to edit or update the mapping in the new environment.

As a rule, when you specify the schema or table in the advanced properties, connection, or object properties, Data Integration honors the attributes in the following order of precedence:

1. **Runtime advanced attributes**. The advanced properties such as schema and table in the Source or Target transformation in a mapping.
2. **Connection attributes**. Attributes such as database and schema set in the connection properties.
3. **Object metadata**. The object selected in the Source or Target transformation in a mapping.

Migration options

When you migrate to a different path, you can choose from one of the following options to update the object path:

Option 1. Update the connection properties to reference the new object

When you import the mapping from, for example, Org 1, into a new organization, for example, Org 2, you can change the existing connection in the **Review Connections** section to map to the connection that has access to the specified schema and table in Org 2.

Option 2. Override the properties from the advanced properties

Before the migration, specify the required schema and table name for the object in the new organization in the advanced properties of the Org 1 mapping.

After the migration, when you run the mapping, the Secure Agent uses the configured advanced parameters to override the object specified in the mapping imported from Org 1.

Option 3. Parameterize the properties in the mapping

You can choose to parameterize the advanced attributes, such as the schema and table name before the migration. You can configure input parameters, in-out parameters, and parameter files in the mapping. When you use a parameter file, you can save the parameter file on a local machine or in a cloud-hosted directory. After you migrate the mapping, do not edit or update the mapping. If you have used in-out parameters for the advanced attributes such as for the schema and table name, you can update these from the parameter file.

Parameterizing only the advanced properties, but not the object

You can choose to parameterize only the advanced properties before the migration, and use them at runtime. After the migration, when you run the mapping, the Secure Agent uses the value of the parameters specified in the advanced properties.

Parameterizing both the advanced properties and the object

When you parameterize both the advanced properties and the object, you can choose to select or deselect the Allow parameter to be overridden at runtime option.

Allow parameter to be overridden at runtime option is selected

When you parameterize both the advanced properties and the object or connection and select the **Allow parameter to be overridden at runtime** option before the migration, you must select a placeholder object in the object properties in the mapping and then specify an override to this placeholder object from the advanced properties. Ensure that the placeholder object contains the same metadata as the corresponding table that you specify as an override. When you run the mapping, the value specified in the advanced property overrides the placeholder object.

Allow parameter to be overridden at runtime option is not selected

When you parameterize both the advanced properties and the object or connection and do not select the **Allow parameter to be overridden at runtime** option before the migration, you do not need a placeholder object and the values specified in the advanced properties take precedence.

Migration restrictions

Consider some rules for some of the configurations before you migrate assets:

- Retain the following transformation, object type, and fields in the advanced properties of a mapping:

Transformations	Object type
Source	Single object, query
Target	Single object
Note: Advanced fields such as the schema and table name are supported for the Source and Target transformations.	

- If you use in-out parameters for the schema and table name and override these parameters from a parameter file, ensure that you define default values for the in-out parameter.

CHAPTER 6

Data type reference

Data Integration uses the following data types in mappings and mapping tasks with Microsoft Fabric Data Warehouse:

Microsoft Fabric Data Warehouse native data types

Microsoft Fabric Data Warehouse data types appear in the Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping in advanced mode.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Microsoft Fabric Data Warehouse and transformation data types

Use only the supported Microsoft Fabric Data Warehouse data types in a mapping or mapping in advanced mode to read from and write to Microsoft Fabric Data Warehouse.

The following table lists the Microsoft Fabric Data Warehouse data types that Data Integration supports and the corresponding transformation data types:

Microsoft Fabric Data Warehouse Data Type	Transformation Data Type	Description
Bigint	Bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
Bit	Integer	1, 0, or NULL
Char	String	1 to 8000 characters
Date	Date/Time	January 1, 1900 to December 31, 9999
Datetime2	Date/Time	1900-01-01 00:00:000000

Microsoft Fabric Data Warehouse Data Type	Transformation Data Type	Description
Decimal	Decimal	Maximum value: 9.999999999999998e+37 Minimum value: -9.999999999999998e+37
Float	Double	Precision 15, scale 0
Int	Integer	-2,147,483,648 to 2,147,483,647
Numeric	Decimal	Precision 1 to 28, scale 0 to 28
Real	Double	Precision 7, scale 0
Smallint	Integer	-32,768 to 32,767
Time	String	00:00:00.000 to 23:59:59.999
Uniqueidentifier	String	36 characters
Varbinary	Binary	1 to 8000 bytes
Varchar	String	1 to 8000 characters

CHAPTER 7

Troubleshooting

Use the following sections to troubleshoot errors in mappings.

Troubleshooting a mapping

The performance of a mapping that uses a Microsoft Fabric Data Warehouse connection is not optimal in the first few runs.

When you run a Microsoft Fabric Data Warehouse mapping, the performance is slow in the first few runs compared to subsequent runs. The slowdown occurs because of a third-party limitation related to the statistics that the query engine creates and optimizes in Microsoft Fabric Data Warehouse. Accurate statistics are crucial for optimal query performance in Microsoft Fabric Data Warehouse, and the initial mapping runs are necessary to generate and optimize these statistics.

For more information on this issue and its resolution, see the Microsoft Fabric Data Warehouse performance guidelines in the Microsoft documentation.

Mapping configured to read or write Date and Datetime data types fails.

A mapping in advanced mode fails in the following cases:

- Data is of the Date data type and the date is less than 1582-10-15.
- Data is of the Datetime data type and the timestamp is less than 1900-01-01T00:00:00Z.

To resolve this issue, specify the following spark session properties in the mapping task or in the custom properties file for the Secure Agent:

- `spark.sql.legacy.timeParserPolicy=LEGACY`
- `spark.sql.parquet.int96RebaseModeInWrite=LEGACY`
- `spark.sql.parquet.datetimeRebaseModeInWrite=LEGACY`
- `spark.sql.parquet.int96RebaseModeInRead=LEGACY`
- `spark.sql.parquet.datetimeRebaseModeInRead=LEGACY`

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